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322840(22)

B. E. (Eighth Semester) Examination, Nov.-Dec. 2019

(New Scheme)

(CSE, IT Engg. Branch)

NEURAL NETWORK and FUZZY LOGIC

Time Allowed : Three hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : *Attempt all questions. Part (a) of each question is compulsory. Attempt any* ***two*** *parts from (b), (c) and (d) of each question.*

Unit-I

1. (a) What is the meaning of Neurophysiology? Define in a few words. 2

(b) Write a major difference between AI & NN and justify your answer using an example for both. 7

(c) Explain Neural Network Architectures. 7

(d) What is ANN? Explain ANN given with a suitable models. 7

Unit-II

2. (a) Define Learning. Why we need of learn for ANN? 2

(b) Explain Supervised & Unsupervised learning with the term of labelled data & unlabelled data. 7

(c) What is Rote Learning? Explain with suitable examples. 7

(d) Explain Hebbian learning with Algorithms. 7

Unit-III

3. (a) Define the role of perceptron in NN models. 2

(b) Explain Adalines & Madalines with its applications. 7

(c) What is XOR problems? How to solve XOR problems. 7

(d) Write a short note on Back Propagation Algorithms. 7

Unit-IV

4. (a) What is Neocgnitron? 2

(b) Explain applications of Pattern Recognition. 7

(c) Distinguish between Speech Generation & Speech Recognition. 7

(d) How to recognize a character in ANN? Explain its steps. 7

Unit-V

5. (a) Define Defuzzification. 2

(b) What is the role of membership value in fuzzy logic? Explain it with suitable examples. 7

(c) Explain Associative Memories with block diagram. 7

(d) What are the main operations & relations of fuzzy sets? Give suitable examples? 7